

Expert Annotation Guidelines for Decision Logic Verification

1 Task Overview and Objectives

This task aims to assess the fidelity of the **CV-DLS (Causal-Vector Decision Logic System)** framework. The objective for the human annotator is to verify whether the structural data (Importance Vectors and Adjacency Matrices) extracted by the framework accurately reflect the decision-making rationales provided by LLM Agents in a simulated data market.

2 Definition of Decision Factors

Annotators evaluate 8 core factors (F_1 – F_8), where all data structures follow this strict order:

- **F1 (Profit Margin):** Increase in unit profit resulting from pricing strategies.
- **F2 (Customer Loyalty):** Consumers' dependence on recommendations or data-sharing benefits.
- **F3 (Competitive Pressure):** Market rivalry driving price adjustments or strategic shifts.
- **F4 (Customer Churn Risk):** The risk of losing customers due to high pricing.
- **F5 (Demand Uncertainty):** Fluctuations in market demand leading to pricing errors.
- **F6 (Net Profit):** The bottom-line trade-off between returns and operational costs.
- **F7 (Profit Maximization):** The primary drive toward maximizing total revenue.
- **F8 (Loss Minimization):** The defensive drive toward mitigating potential risks or losses.

3 Audit Dimensions

Annotators compare the **Original Rationale Text** with the **Automated Extraction Results** based on:

1. **Importance Vector:** A vector of length 8 representing factor weights (1 = Primary, 2 = Secondary, 3 = Irrelevant).
2. **Adjacency Matrix:** An 8×8 binary matrix where $M[i][j] = 1$ indicates a logical connection (causality or trade-off) between F_i and F_j .

4 Judgment Criteria and Semantic Tolerance

Annotators provide a binary judgment: **1 (Valid)** or **0 (Invalid)**.

4.1 Criteria for Outputting “1” (Valid)

- **Semantic Alignment:** The core logic of the extraction matches the general meaning of the text.
- **Reasonable Inference:** Factors can be logically inferred from economic common sense even if not explicitly named.
- **Non-Fatal Variance:** Minor subjective differences in importance levels are acceptable if the direction of influence is correct.
- **Structural Consistency:** The matrix captures the 2–3 most critical logical chains.

4.2 Criteria for Outputting “0” (Invalid)

- **Factual Contradiction:** The extraction highlights a factor that the text explicitly ignores.
- **Logical Absence:** The central causal chain of the rationale is entirely missing.
- **Spurious Correlation:** The matrix contains links that have no basis in the text.

5 Illustrative Case Study

Sample Rationale	System Output	Judgment	Reasoning
”Due to a competitor’s price cut, we lowered pricing to prevent user churn.”	$F_3 = 1, F_4 = 1$; Matrix: $F_3 \rightarrow F_4$	1	Accurate capture of ”Competition $\rightarrow$ Churn.”